

SAI Performance Analyst

Day-Before Study Guide

Physiotherapy — compact revision notes

Includes A1, A2 and Section B weak-area notes, formulas, terminology, tables, memory aids and self-tests.

Prepared from the repository study notes. Current A2 snapshot: 2026-08-21.

Performance Analyst — day-before study notes

These notes are deliberately compact: revise the bold terms, redraw the small diagrams, and answer the “self-test” prompts without looking. They are aligned to the SAI notification and the weak-area labels used by the question bank. They are revision aids, not substitutes for clinical judgement or the full reference texts.

Abbreviations are expanded at first use. The quick abbreviation glossary is available when a short form still feels unfamiliar.

How to read the diagrams

- → means “then” or “leads to.” It does not always mean that the first item caused the second item.
- + means “together with.”
- / means “or,” where there are alternatives.
- In a timeline, read the years in increasing order. Each date is a separate fact unless the note explicitly says that one event caused another.

Suggested one-day route

Block	Notes	What to do
1	A1 core, testing	Memorise definitions and formulas
2	A1 technology/statistics, analytics	Rework every formula once
3	A1 performance/recovery, holistic/travel/ethics	Convert each checklist into a scenario answer
4	A2 Olympics/Asian Games, Indian ecosystem	Recall the institution–scheme map
5	A2 anti-doping, championships, governance	Use the “do not confuse” boxes
6	B notes 1–3	Draw biomechanics, anatomy and energy-system charts
7	B notes 4–6	Recall modality parameters, tests and reasoning sequence
8	B notes 7–9	Practise emergency/RTP/research decision trees

Exam-use reminders

- The official CBT is **100 questions in 120 minutes**, with A1/A2/B/C weighted **32/8/40/20** and negative marking of one-quarter of the question mark for an incorrect answer.
- In a clinical stem, answer in this order: **first check danger; then form a working diagnosis; protect the athlete; reassess objectively; and progress only when criteria are met.**

- If two options look plausible, prefer the one that is **criteria-based, athlete-centred, evidence-informed and safe**, not the one that promises a fixed timeline.
- Recheck current A2 facts against the official source linked in the note. Current snapshot in these notes: **2026-08-21**.

Provenance

Source IDs in each note refer to the repository catalog at `src/data/sourceCatalog.ts`. Local books are in `books/`, and the SAI notification is `_Final_ NOTIFICATION FOR PA REGULAR RECRUITMENT_30.01.2026.pdf`.

Quick abbreviation glossary

The topic notes spell these out at first use. Use this page only as a quick lookup.

A1 and A2

Short form	Meaning
BMI	Body mass index
ATP / PCr	Adenosine triphosphate / phosphocreatine
GPS / IMU	Global positioning system / inertial measurement unit
HRV	Heart-rate variability
EMG	Electromyography
RPE	Rating of perceived exertion
LSI	Limb symmetry index
ICC / SEM / MDC	Intraclass correlation coefficient / standard error of measurement / minimum detectable change
CV / CI	Coefficient of variation / confidence interval
IOC / IF / NOC	International Olympic Committee / International Federation / National Olympic Committee
IOA / OCA	Indian Olympic Association / Olympic Council of Asia
MYAS / SAI	Ministry of Youth Affairs and Sports / Sports Authority of India
TOPS / NSDF / NCOE / NSF	Target Olympic Podium Scheme / National Sports Development Fund / National Centre of Excellence / National Sports Federation
WADA / NADA	World Anti-Doping Agency / National Anti-Doping Agency India
TUE	Therapeutic use exemption
POSH	Prevention of Sexual Harassment of Women at Workplace
FIFA / ICC / FIBA	Football / cricket / basketball international governing bodies
ODI / T20	One-day international / Twenty20 cricket

B — physiotherapy

Short form	Meaning
ROM / MMT	Range of motion / manual muscle testing
ACL	Anterior cruciate ligament
ATFL / CFL	Anterior talofibular ligament / calcaneofibular ligament
FABER	Flexion–abduction–external rotation test
TENS / NMES / IFC	Transcutaneous electrical nerve stimulation / neuromuscular electrical stimulation / interferential current
SWD / ESWT	Short-wave diathermy / extracorporeal shock-wave therapy

Short form	Meaning
PNF	Proprioceptive neuromuscular facilitation
SCAT6	Sport Concussion Assessment Tool 6
AED / CPR	Automated external defibrillator / cardiopulmonary resuscitation
RED-S	Relative energy deficiency in sport
RCT	Randomised controlled trial
PICO	Population, intervention, comparison, outcome
RR / ARR / NNT / OR	Relative risk / absolute risk reduction / number needed to treat / odds ratio

A1 — Core Disciplines Foundational Concepts

The one-minute map

1. Measure the athlete.
2. Explain the physiology.
3. Analyse movement.
4. Identify the main limiting factor.
5. Train and recover.
6. Protect long-term wellbeing.

The eight foundational disciplines are: **anthropometry, exercise physiology, physiotherapy, biomechanics, biochemistry, strength and conditioning, psychology, and nutrition**. Read them as one complete athlete-support team, not as isolated subjects.

High-yield facts

Area	Remember
Anthropometry	Body mass index (BMI) = mass (kg) / height² (m²) . It is a size index, not a direct body-fat measure; muscular athletes can be misclassified. Skinfolds, girths, dual-energy X-ray absorptiometry or other methods answer different body-composition questions.
Exercise physiology	The adenosine triphosphate–phosphocreatine system (ATP–PCr) gives the fastest power for very brief maximal work; glycolysis contributes strongly to high-intensity efforts; oxidative metabolism dominates sustained work. The systems overlap—do not memorise rigid “switch” times.
Physiotherapy	Rehabilitation is a staged exposure to load: protect the injured tissue, maintain capacity elsewhere, restore specific function, then expose the athlete to sport demands.
Biomechanics	Force changes linear motion; torque = force × perpendicular moment arm changes rotation. A longer moment arm increases torque for the same force.
Biochemistry	Adenosine triphosphate (ATP) is the immediate energy currency. Glycogen supports high-intensity work; protein provides amino acids for repair/adaptation but is not the primary rapid fuel for maximal sprinting.
Strength & conditioning	Apply specificity, overload, progression, individualisation, variation and recovery . Training adaptation is specific to the stimulus and dose.
Psychology	Arousal and performance often follow an inverted-U pattern, but the optimal zone is individual and task-dependent. Self-efficacy affects effort, persistence and response to setbacks.
Nutrition	Carbohydrate supports high-intensity training; protein supports repair/adaptation; fat is essential for energy and cell functions. Approximate energy values: carbohydrate 4 kcal/g, protein 4 kcal/g, fat 9 kcal/g.

Energy-system picture

Energy system	Main strength	Typical exam example
Adenosine triphosphate–phosphocreatine	Fastest energy supply, but small capacity	Very short maximal burst
Glycolytic	High energy-supply rate, with metabolite accumulation	A hard effort lasting longer than the first burst

Energy system	Main strength	Typical exam example
Oxidative	Slower supply rate, but largest capacity	Sustained work and recovery between efforts

All three systems work at the same time. The question is which one contributes most, not which one is switched on.

Exam distinction: "primary" does not mean "exclusive." A 30-second effort uses all systems, with different relative contributions.

Fast movement logic

1. Identify the task and external demand.
2. Identify the force, direction and lever arm.
3. Ask which tissue/system limits the task.
4. Change the training variable that targets that limiter.
5. Re-test the same meaningful outcome.

Common traps

- BMI is not body-fat percentage.
- Lactate is not simply "the waste product"; it is a usable metabolic intermediate, while associated acidity and other factors contribute to fatigue.
- Flexibility, mobility, stability, strength and power are not interchangeable.
- More training is not automatically better: adaptation requires an appropriate **stimulus–recovery balance**.
- Correlation between a biomarker and performance does not prove that changing the biomarker will improve performance.

Easy ways to remember it

- Say the eight disciplines aloud in this order: **anthropometry, exercise physiology, physiotherapy, biomechanics, biochemistry, strength and conditioning, psychology, nutrition**.
- In an applied question, ask three plain questions: **What force is involved? Which tissue or body system receives it? What is limiting the athlete?**
- For food questions, remember the simple rule: **carbohydrate mainly fuels hard work, protein supports repair, and fat supports energy and body functions**.

Self-test

1. Why can a rugby forward have a high BMI without high adiposity?
2. Which system supplies the fastest ATP resynthesis, and why is it not the only system used?
3. If the same force is applied farther from a joint centre, what happens to torque?
4. Why should a rehabilitation exercise eventually resemble the athlete's sport task?

Sources

sai-notification-2026; basic-biomechanics-hall (Hall, Chs. 1–6); physiology-sport-exercise-kenney; nsca-essentials-5; clinical-sports-nutrition-burke; brukner-khan-clinical-sports-medicine; kisner-therapeutic-exercise.

A1 — Testing Parameters, Standardization & Athlete Assessment

Measurement vocabulary

Term	Meaning	Memory hook
Validity	Measures what it claims to measure	V = target
Reliability	Repeats consistently under similar conditions	R = repeat
Objectivity	Little dependence on who administers/scored it	O = observer-proof
Sensitivity	True-positive rate; detects people who have the condition	Sensitive = catches
Specificity	True-negative rate; excludes people who do not have the condition	Specific = clears
Responsiveness	Detects meaningful change over time	Response = change
Norm-referenced	Compared with a reference population	"How do you compare?"
Criterion-referenced	Compared with a meaningful standard/threshold	"Did you meet the target?"

The formulas worth writing once

In these formulas, **TP** means true positive, **TN** true negative, **FP** false positive, and **FN** false negative. **PPV** means positive predictive value and **NPV** negative predictive value. **SD** means standard deviation; **ICC** means intraclass correlation coefficient; **SEM** means standard error of measurement; **MDC** means the smallest change likely to be real; and **LSI** means limb symmetry index. **CV** means coefficient of variation; **z** means standard score; and **MDC95** means the minimum detectable change at 95% confidence.

$\text{Sensitivity} = \text{TP} / (\text{TP} + \text{FN})$	$\text{Specificity} = \text{TN} / (\text{TN} + \text{FP})$
$\text{PPV} = \text{TP} / (\text{TP} + \text{FP})$	$\text{NPV} = \text{TN} / (\text{TN} + \text{FN})$
$\text{CV}\% = (\text{SD} / \text{mean}) \times 100$	$z = (\text{individual} - \text{reference mean}) / \text{reference SD}$
$\text{SEM} = \text{SD} \times \sqrt{1 - \text{ICC}}$	$\text{MDC95} = 1.96 \times \sqrt{2} \times \text{SEM}$
$\text{LSI}\% = \text{involved side} / \text{uninvolved side} \times 100$	

Interpretation: a negative result from a very sensitive test is useful for ruling out a condition; a positive result from a very specific test is useful for ruling it in. Predictive values change with prevalence.

Standardisation sequence

Use this sequence for a safe, repeatable test:

1. State the purpose of the test.
2. Screen safety and obtain informed consent.
3. Familiarise the athlete with the task.
4. Calibrate the equipment.
5. Standardise preparation and the testing environment.
6. Administer the test and record the raw data.
7. Check data quality and compare it with the correct reference.
8. Make the training decision and schedule a repeat test.

Control before the athlete arrives

- Same time of day where practical; record sleep, food, caffeine, hydration, medication, recent training and illness.
- Same warm-up, instructions, equipment, surface, footwear, tester and rest intervals.
- Calibrate instruments and record serial number/software version when relevant.
- Use the same order of tests; fatigue can contaminate later results.
- Define stopping criteria, emergency plan and contraindications before testing.
- Obtain informed consent; explain purpose, risks, expected discomfort, data use and the right to stop.

Choose a battery by question, not by habit

Choose a test in five steps: **identify the practical question, decide what ability or quality must be measured, choose a valid test, use a reliable protocol, and make sure the result can change a decision.**

Example: "Why is repeated sprint output falling?" may require sprint time, recovery heart rate, **rating of perceived exertion (RPE)**, sleep/travel and training-load context—not a single laboratory value.

Lab versus field

Laboratory	Field
More control and often greater measurement precision	More sport-specific and ecologically valid
Costly, less accessible, sometimes less representative	More affected by weather, surface and tester variation
Useful for mechanism and precise comparison	Useful for repeated monitoring and decision-making

Common traps

- A reliable test can be reliably wrong; reliability does not guarantee validity.
- A statistically significant change smaller than the **minimum detectable change (MDC)** or the **minimum clinically important difference (MCID)** may not be actionable.
- "Gold standard" is not automatically the best field test for every athlete.
- Never compare scores collected under materially different protocols as if they were the same measurement.

Easy ways to remember it

- **Validity asks "does it measure the right thing?" Reliability asks "would it repeat?"**
- A negative result from a very sensitive test is useful for ruling out a condition. A positive result from a very specific test is useful for ruling it in.
- Before every test, check five things in plain language: **Is the athlete safe? Is the athlete prepared? Is the method fixed? Is the equipment calibrated? Is the environment recorded?**

Self-test

1. Which metric answers "how repeatable is this test?"
2. Why can a low-sensitivity negative test be falsely reassuring?
3. What is the practical difference between SEM and MDC?
4. Name five conditions that should remain constant in a longitudinal test.

Sources

sai-notification-2026 ; acsm-getp-12 ; nsca-essentials-5 ; magee-orthopaedic-assessment ; world-physiotherapy-consent-2023 ; world-physiotherapy-ethics-2023 .

A1 — Emerging Technologies & Biostatistics

Match the tool to the question

Tool	Main output	High-yield limitation
Global positioning or local positioning system (GPS/LPS)	Distance, speed, acceleration, high-speed running	Sampling rate, satellite/venue signal and device validity affect short bursts
Inertial measurement unit (IMU)	Acceleration, angular velocity, orientation, impacts	Placement, drift and algorithm differences
Force plate	Ground-reaction force, impulse, rate of force development, asymmetry	Expensive; a single jump is not the whole athlete
Motion capture/video	Joint angles, timing, technique	Marker/occlusion or camera-angle error; 2-D can hide 3-D motion
Heart rate and heart-rate variability (HR/HRV)	Cardiovascular response/autonomic proxy	Strongly affected by sleep, illness, posture, breathing and measurement conditions
Electromyography (EMG)	Electrical muscle activity timing/amplitude proxy	Crosstalk, electrode placement and normalisation matter
Athlete-management platform	Merges wellness, load, injury and availability data	Garbage in; privacy/access control; dashboards can create false certainty
Artificial intelligence and machine learning (AI/ML)	Classification, prediction, pattern detection	Needs representative data, external validation and explainability

Descriptive statistics

Mean = $\Sigma x / n$ Range = maximum - minimum
Sample variance = $\Sigma(x - \text{mean})^2 / (n-1)$ Standard deviation (SD) = $\sqrt{\text{variance}}$
Coefficient of variation (CV%) = $\text{SD} / \text{mean} \times 100$
Standard score (z) = $(x - \text{reference mean}) / \text{reference SD}$

- **Mean:** sensitive to outliers; useful for roughly symmetric data.
- **Median:** robust to skew/outliers; report the interquartile range (IQR) with it.
- **Standard deviation:** spread around the mean; not the same as standard error.
- **Confidence interval (CI):** plausible range for a population estimate, not the probability that a fixed parameter is inside the interval.

Inferential/statistical literacy

Concept	What it actually says
p-value	How unusual the data would be under the null model; not the probability the hypothesis is true
95% confidence interval (CI)	Precision/uncertainty around an estimate; check whether it crosses a meaningful threshold
Effect size	Magnitude of a difference/relationship; pair with uncertainty and context
Type I error	False positive; reject a true null

Concept	What it actually says
Type II error	False negative; miss a real effect
Correlation	Association, not causation
Practical significance	Whether the change matters to the athlete, not merely whether $p < 0.05$

Choosing a visual

If you want to show...	Use...
Change over time	A line chart
Differences between groups	A dot plot or box plot
How values are distributed	A histogram
The relationship between two variables	A scatter plot
Parts that make up a whole	A stacked bar chart
Agreement between two measurement methods	A Bland–Altman plot

Data-quality checklist

For data quality, follow this checklist: **define the variable, check the units, inspect missing values and unusual values, verify the sensor and test protocol, decide the analysis before looking for a preferred result, show the uncertainty, explain the practical decision, and protect privacy.**

Do not confuse: accuracy (closeness to truth), precision (repeatability), validity (construct/criterion meaning), and reliability (consistency).

Easy ways to remember it

- Match the tool to its job: **positioning systems track travel, inertial units track movement, force plates measure pushes, cameras show angles, and electromyography shows muscle electrical activity.**
- For statistics, remember three questions: **Where is the centre? How spread out are the values? How uncertain is the estimate?**
- When you see “statistically significant,” immediately ask: **How large is the change? How certain is it? Does it matter to this athlete?**

Self-test

1. Which sensor is most directly suited to measuring impulse?
2. When would median/IQR be preferable to mean/SD?
3. Why is a model with excellent training accuracy not automatically useful in a new team?
4. What is the difference between statistical and practical significance?

Sources

sai-notification-2026; nsca-essentials-5; gps-validity-scott-2016; wearable-inertial-camomilla-2018; wearable-athlete-sensors-seshadri-2019; world-physiotherapy-ebp-2023.

A1 — Data Interpretation & Applied Analytics

The performance-analyst loop

Use this decision cycle:

1. Ask a clear performance question.
2. Establish a baseline.
3. Collect comparable data.
4. Check and clean the data.
5. Display the pattern.
6. Interpret it with uncertainty.
7. Communicate the finding and intervention.
8. Re-test to see whether the decision helped.

The output is not a dashboard; it is a **defensible decision**.

Useful calculations

```
Absolute change = post - pre
% change = (post - pre) / pre × 100
Training monotony = weekly mean daily load / standard deviation of daily load
Training strain = weekly load × monotony
Session load = session duration (min) × rating of perceived exertion
Limb symmetry index (LSI) = involved / uninvolved × 100
```

Use acute:chronic workload ratios cautiously. They are summaries of load history, not a universal injury threshold or a substitute for athlete context.

Read a performance profile

For each athlete, record four things in a performance profile:

Area	Ask
Strengths	What can the athlete already do well?
Limiters	Which important, changeable weakness is restricting performance?
Recovery and health	Is the athlete available and recovering sufficiently?
Speed and sport demands	Which qualities must transfer to the actual sport task?

For each domain record: **current level, target, confidence in measurement, trend, limiting factor, action and review date**. A low score is not automatically the limiter; a high-impact, modifiable deficit is more actionable.

Longitudinal interpretation

- Compare like with like: same test, protocol, timing, environment and athlete state.
- Plot individual data, not only team averages; averages can hide responders and non-responders.
- Separate **signal** from **noise** using reliability, measurement error, the minimum detectable change and repeated observations.
- Mark illness, travel, menstrual-cycle context where voluntarily and appropriately shared, injury and major training changes on the timeline.
- Use thresholds as prompts for review, not as automatic diagnoses.

Analytics decisions

Pattern	First question	Sensible action
Performance down, wellness normal	Test error? fatigue from unrecorded load?	Recheck protocol and training exposure
Performance down, wellness down	Recovery/illness/travel?	Reduce or modify load; screen clinically
One sensor abnormal	Device/placement/algorithm?	Cross-check with another measure
Group average stable, athlete variable	Who is hidden by the mean?	Individualise monitoring
Strong correlation	Is the relationship causal and useful?	Examine timing, confounding and intervention evidence

Communication template

Report every finding in this order: **what changed, how confident you are, why it matters to performance, what you recommend, and when you will review it.**

Example: "Repeated-sprint time is 4% slower than the athlete's reliable baseline across two sessions; sleep and recent high-speed volume are also reduced. Treat this as a recovery flag, modify today's high-speed exposure, and reassess after recovery."

Easy ways to remember it

- For every result, work through five plain questions: **What is the context? Am I comparing like with like? Could this be measurement error? What action follows? When will I check again?**
- Every chart must answer: **What changed? Compared with what? How sure are we? What will we do?**

Self-test

1. Calculate session load for 60 minutes at a rating of perceived exertion of 7.
2. Why is one abnormal wearable value insufficient for a training decision?
3. What fields belong in a performance profile besides the score?
4. Why should an acute:chronic ratio never be treated as a diagnosis?

Sources

sai-notification-2026; nsca-essentials-5; ioc-load-illness-2016; ioc-load-management-consensus-2016; gps-validity-scott-2016; world-physiotherapy-ebp-2023.

A1 — Performance Enhancement & Recovery Protocols

Programme logic

Build a programme in this order: **set the goal, analyse the sport demand, measure a baseline, prescribe the training dose, plan recovery, monitor the response, adjust the dose, and expose the athlete to the sport task.**

Set the dose using **frequency, intensity, time and type**. Add volume, density, complexity and velocity when appropriate.

Training principles

Principle	Exam meaning
Specificity	Adaptation follows the energy system, movement, velocity and decision demand trained
Overload	Stimulus exceeds the athlete's current habitual demand
Progression	Increase challenge gradually and deliberately
Individualisation	Dose depends on athlete, sport, position, history and readiness
Variation/periodisation	Organise stress and recovery across time; avoid random change
Reversibility	Unused adaptations decline with detraining
Recovery	Adaptation is built between exposures, not during unbroken stress

Strength versus power

Strength = force capacity
Power = force × velocity

An athlete can be strong but slow. To improve power, preserve force while training intent and movement velocity; select loads and exercises that fit the task.

Recovery hierarchy

1. Fix the dose: training, competition and travel load
2. Sleep and circadian routine
3. Fuel and hydration
4. Active recovery / mobility / psychological down-regulation
5. Targeted modalities when a clear purpose and tolerable evidence exist

Tool	Use carefully
Active recovery	Low-intensity movement may reduce perceived stiffness; it is not a cure for excessive load
Cryotherapy/heat	Context-specific effects; do not use a modality to mask danger or replace diagnosis
Compression	May help symptoms/recovery perception in some settings; individual response varies
Massage	Short-term symptoms/relaxation; does not replace progressive loading

Tool	Use carefully
Sleep	Protect duration, regularity, dark/cool room and pre-sleep routine; individual need varies

Injury-risk lens

The risk or readiness decision should consider five pieces of information: **the work completed, the athlete's internal response, tissue capacity, recovery, and context such as illness, travel or heat.**

Progress load by response and criteria, not calendar alone. Watch spikes, monotonous training, sudden high-speed exposure, illness, low energy availability and poor sleep.

Nutrition and supplements

- Carbohydrate: high-intensity training and glycogen restoration.
- Protein: distribute adequate high-quality protein across the day; increase attention during immobilisation/rehabilitation.
- Hydration: individualise to sweat rate, climate, duration and sodium losses; avoid both large dehydration and overdrinking.
- For supplements, check these in order: **food first, evidence, contamination risk, athlete-specific need and third-party testing.** No supplement is automatically safe because it is "natural." The **World Anti-Doping Agency (WADA)** rules still apply.

Easy ways to remember it

- A good programme answers four questions: **Is it specific? Is it difficult enough? Is it progressed gradually? Is recovery planned?**
- Put recovery in this order: **correct the training dose, protect sleep, provide food and fluids, then consider extra recovery tools.**
- Power requires both **force and speed.** A plan that trains only force is incomplete when the sport requires fast movement.

Self-test

1. Why is progressive loading preferable to an arbitrary "week 6 clearance"?
2. Distinguish strength from power in one sentence.
3. Name the first three recovery levers before passive modalities.
4. Why is a supplement a risk-management question as well as a performance question?

Sources

sai-notification-2026; nsca-essentials-5; brukner-khan-clinical-sports-medicine; sports-injury-prevention-rehabilitation-joyce-lewindon; ioc-load-illness-2016; ioc-load-management-consensus-2016; ioc-reds-2023; ioc-sports-nutrition-consensus; wada-prohibited-list-2026.

A1 — Holistic Development, Travel & Ethics

The athlete is a system

An athlete’s availability and long-term performance depend on several areas:

Area	Why it matters
Physical capacity	Determines what the athlete can do
Mental health	Affects motivation, confidence, concentration and recovery
Fuel and energy	Supports training, healing and normal body functions
Sleep and travel	Affect alertness, adaptation and readiness
Relationships and ethics	Affect safety, trust and sustainable participation

Use a **biopsychosocial** lens: a performance change may be physical, psychological, social, environmental—or several at once.

High-yield wellbeing concepts

Term	Distinguish it from
Overreaching	Short-term planned/unplanned performance dip; may be functional when followed by recovery
Overtraining syndrome	Persistent maladaptation with prolonged performance decline and broader symptoms; diagnosis is not made from one bad session
Burnout	Exhaustion, reduced accomplishment and sport devaluation; not simply ordinary fatigue
Relative energy deficiency in sport (RED-S)	Health/performance consequences of low energy availability affecting multiple physiological systems; can affect any sex
Long-term athlete development (LTAD)	Development is long-term and stage-sensitive; avoid adult training assumptions for growing athletes

Travel and environmental readiness

Travel stage	Main actions
Before departure	Plan time-zone adjustment, sleep, food, fluids, medicines and emergency contacts
During travel	Move regularly, hydrate, protect sleep, manage meals and reduce infection exposure
After arrival	Set a local sleep/light routine, monitor symptoms and reintroduce hard training gradually

- **Eastward travel** usually requires advancing the body clock; **westward** travel usually requires delaying it. The direction of light exposure matters; do not give blanket “bright light at any time” advice.
- Shift gradually where possible; use a consistent sleep/wake anchor, morning/evening light strategically, and avoid alcohol or unprescribed sedatives.
- At altitude, hypobaric hypoxia increases ventilatory and cardiovascular stress. Headache, nausea, dizziness and sleep disturbance require monitoring; worsening neurological or respiratory symptoms are red flags.
- In heat/humidity, acclimatisation, work–rest planning, cooling and individual hydration strategy matter. Suspected exertional heat stroke is an emergency, not a “push through” problem.

Ethics checklist

Ethical practice means **obtaining consent, working within competence, keeping information confidential, managing conflicts of interest, supporting clean sport, safeguarding the athlete, keeping accurate records, and escalating a serious concern.**

- Share only necessary health data with the appropriate team member.
- Explain risks, benefits, alternatives and data use; consent is voluntary and can be withdrawn.
- Record facts, not speculation; correct errors transparently.
- Disclose conflicts of interest and do not let selection pressure override welfare.
- The athlete-support team educates about anti-doping and **therapeutic use exemptions (TUEs)**; it does not prescribe shortcuts.

Easy ways to remember travel and ethics

Before travel, prepare a **sleep plan, food and fluids, emergency contacts, medicines and consent for information sharing**. During travel, protect sleep, move regularly and record important changes. After arrival, adjust light and sleep gradually, then reintroduce hard training only when the athlete is ready.

For ethical questions, choose the response that protects **safety, autonomy, privacy, accurate records and fair treatment**.

Self-test

1. How does eastward jet lag differ from westward jet lag?
2. Why is a single poor performance not enough to diagnose overtraining syndrome?
3. What is the ethical difference between sharing necessary availability information and sharing a full medical record?
4. Name three immediate priorities when travel and competition overlap.

Sources

sai-notification-2026; ioc-load-illness-2016; ioc-load-management-consensus-2016; ioc-reds-2023; brukner-khan-clinical-sports-medicine-vol2; world-physiotherapy-ethics-2023; world-physiotherapy-consent-2023; wada-istue-2023.

A2 — Olympic & Asian Games

Institution map

Organisation	Main responsibility
International Olympic Committee	Governs the Olympic Movement and protects the Olympic Charter
International Federation	Makes the technical rules for one sport
Indian Olympic Association	Represents India as its National Olympic Committee
National Sports Federation	Develops and administers one sport in India
Host organising committee	Delivers one particular edition of the Games

Do not confuse the **International Olympic Committee** (Olympic Movement governance) with an **International Federation** (technical sport rules), a **National Olympic Committee** (national representation) or a host organising committee (delivery of one Games).

Core timeline

Fact	Recall
International Olympic Committee constituted	Paris Congress, 1894; modern Olympic Games began at Athens in 1896
Olympiad	Four consecutive calendar years; the Games need not be held in every year of the Olympiad
Olympic motto	Citius, Altius, Fortius — Communiter: Faster, Higher, Stronger — Together
Olympic symbol	Five interlaced rings express the union of the Olympic continents/athletes and the meeting of the world through sport; do not assign one official ring to one continent
Olympic Charter	Fundamental principles, rules and bye-laws governing the Olympic Movement
Asian Games	First held in New Delhi, 1951 ; continental multi-sport Games under the Olympic Council of Asia
Aichi-Nagoya 2026	20th Asian Games , Japan; 19 September–4 October 2026 (current official schedule)
Next Summer Olympic hosts	Paris 2024 was followed by Los Angeles 2028 and Brisbane 2032

India-related facts to learn

Learn these as four separate facts in time order:

1. **1894 — Paris:** the International Olympic Committee was created at the Paris Congress.
2. **1896 — Athens:** the first modern Olympic Games were held.
3. **1951 — New Delhi:** the first Asian Games were held.
4. **2008 — Beijing:** Abhinav Bindra won India's first individual Olympic gold medal, in the 10-metre air-rifle event.

The years are the important part. The arrows in the earlier version were only meant to connect these facts; they did not represent one event causing the next.

How to answer institution questions

- If the question asks who writes the technical rules for a sport, choose the **International Federation**.
- If it asks who represents India in the Olympic Movement, choose the **Indian Olympic Association**, India's National Olympic Committee.
- If it asks who delivers one edition of the Games, choose the **host organising committee**.
- If it asks about principles, ownership, the symbol or the motto, look for the **Olympic Charter or International Olympic Committee**.
- For a host-city question, read the year carefully: a scheduled event, a current event and a completed event are different statements.

Easy ways to remember it

Remember the order as a sentence: **the International Federation makes the sport rules, the National Olympic Committee represents the country, and the host committee runs that edition.**

Keep the three dates together: **1894 (International Olympic Committee), 1896 (first modern Olympic Games), 1951 (first Asian Games).**

Self-test

1. What is the difference between an Olympiad and an Olympic Games edition?
2. Which body is responsible for sport-specific technical rules?
3. Where and when were the first Asian Games held?
4. What does "Communiter" add to the Olympic motto?

Sources

sai-notification-2026; ioc-olympic-charter-2025; olympics-official; asian-games-official; oca-games-timeline; oca-aichi-nagoya-2026; ioa-organisation-official.

A2 — Indian Sports Ecosystem & Government Schemes

Institution–scheme map

Organisation or programme	Main role
Ministry of Youth Affairs and Sports	National policy, schemes and government oversight
Sports Authority of India	Programme implementation, centres, coaching, sports science and athlete support
Target Olympic Podium Scheme and Mission Olympic Cell	Targeted support for elite athletes with major-Games potential
Khelo India	Participation, competitions, talent development and facilities
National Sports Development Fund	Funding and resources for sport and elite preparation
Indian Olympic Association	Olympic representation for India
National Sports Federation	Administration and development of one sport
National Centres of Excellence	High-performance training with integrated sports-science support

Scheme flash table

Name	Purpose / recognition cue
Khelo India	Broad-based development: sports participation, competitions, talent identification, infrastructure, centres/academies, inclusion and community access
Target Olympic Podium Scheme (TOPS)	Targeted support for athletes with realistic Olympic/Paralympic/major-Games potential; Mission Olympic Cell and technical support are central cues
National Sports Development Fund (NSDF)	Mobilises financial support for sports development and high-performance needs such as training, travel and equipment
Sports Authority of India (SAI)	Implements athlete-development and high-performance programmes; provides coaching, facilities, sports science, medical/rehabilitation and support services
National Centres of Excellence (NCOEs)	National high-performance centres with sport-specific training and multidisciplinary support—coaches, sports science, medicine, nutrition and allied services
Indian Olympic Association (IOA)	India's National Olympic Committee; national representation and Olympic Movement interface
National Sports Federation (NSF)	Governs and develops a particular sport in India; rules, selection and competition administration are sport-specific

How to solve scheme questions

Problem described in the question	Most likely answer
Participation, facilities or a talent pipeline	Khelo India
Elite medal pathway or individual support	Target Olympic Podium Scheme
High-performance campus and integrated support	National Centre of Excellence / Sports Authority of India
Funding or resource mobilisation	National Sports Development Fund
National Olympic representation	Indian Olympic Association
Sport-specific selection or technical rules	National Sports Federation

Governance distinctions

- The **Ministry of Youth Affairs and Sports** is the policy/administrative ministry; it is not the same as the **Sports Authority of India**.
- The **Sports Authority of India** is an implementing and athlete-support organisation; do not reduce it to only coaching.
- A **scheme** is not a constitutional sports body.
- A recognised **National Sports Federation** is not automatically the **Indian Olympic Association** and does not replace the International Olympic Committee or an International Federation.
- Check the scheme's official operational period and current guidelines; scheme components can be revised.

Easy ways to remember it

- **Khelo India is broad, TOPS is targeted, an NCOE is a high-performance centre, and the National Sports Development Fund is a funding route.**
- **The Sports Authority of India implements programmes, the Indian Olympic Association represents India, and a National Sports Federation runs one sport.**

Self-test

1. A district needs facilities and talent identification. Which scheme family is the closest fit?
2. An Olympic medal prospect needs targeted international preparation. Which programme is the likely cue?
3. Who is the national Olympic representative body, and who governs a particular sport?
4. Why is an NCOE different from a generic local sports facility?

Sources

sai-notification-2026 ; myas-khelo-india-2025 ; myas-khelo-india-revised-guidelines-2025 ; myas-annual-report-2024 ; sai-tops-official-2026 ; myas-nsdf-official-2026 ; sai-about-official-2026 ; sai-ncoe-official-2026 ; ioa-organisation-official ; myas-recognised-nsf-2025 .

A2 — Anti-Doping Regulations & Clean Sport

The clean-sport chain

Clean sport has four parts:

1. Know the current Prohibited List and the World Anti-Doping Code.
2. Check medicines and supplements, and apply for a therapeutic use exemption when the criteria are met.
3. Follow testing, whereabouts and sample-integrity requirements.
4. Understand results management, hearings and appeals.

The **2026 World Anti-Doping Agency (WADA) Prohibited List took effect on 1 January 2026**. Always distinguish the current List from the broader Code and from Indian implementation through the **National Anti-Doping Agency India (NADA India)**.

The 11 anti-doping rule violations

Article	Cue
2.1	Presence of a prohibited substance/metabolite/marker in a sample
2.2	Use or attempted use
2.3	Evading, refusing or failing to submit to sample collection
2.4	Whereabouts failure by a whereabouts athlete
2.5	Tampering or attempted tampering
2.6	Possession
2.7	Trafficking or attempted trafficking
2.8	Administration or attempted administration
2.9	Complicity or attempted complicity
2.10	Prohibited association
2.11	Discouraging/reporting retaliation or other prohibited retaliation

To remember the violations, group them instead of memorising a long letter string: **what was found in the sample; what the athlete or support person did; possession, trafficking or association; and failures in the testing process.**

Rules that appear in stems

- **Strict liability:** for presence, the athlete is responsible for what is found in the sample; intent, fault or negligence is not required to establish the basic presence violation. Sanction/defence questions are separate.
- **Athlete responsibility:** check medicines and supplements, disclose relevant use, follow testing procedures, keep whereabouts where applicable and ask before acting when unsure.
- **Support-person responsibility:** do not administer, traffic, conceal, pressure, tamper or retaliate.
- **Supplement risk:** "natural," over-the-counter or trusted by a teammate does not mean contamination-free.

Therapeutic use exemption (TUE): four-part decision test

For a therapeutic use exemption, ask four questions:

1. Does the athlete have a genuine medical condition?
2. Is the treatment needed to restore normal health?
3. Is there no reasonable permitted alternative?
4. Will the treatment not improve performance beyond normal health?

Emergency/retroactive therapeutic use exemption rules are narrow and documentation-dependent. A TUE is not a general permission slip; verify the current National Anti-Doping Agency India/World Anti-Doping Agency process before treatment.

Prohibited-list categories

- **S0** non-approved substances.
- **S1** anabolic agents; **S2** peptide hormones/growth factors; **S3** beta-2 agonists; **S4** hormone/metabolic modulators; **S5** diuretics/masking agents.
- **S6** stimulants; **S7** narcotics; **S8** cannabinoids; **S9** glucocorticoids.
- **M1–M3** prohibited methods; **P1** beta-blockers in particular sports.
- "In-competition only" is not the same as "always prohibited"; check route, dose, sport and formulation.

Easy ways to remember it

- For any medicine or supplement, do three things: **check the current list, check the route and competition status, and look up the exact product.**
- Keep a written record of the **medical reason, source check, exemption decision if needed and final treatment decision.**
- For a question, first decide whether it is about **what was found in a sample, what someone did, possession or association, or the testing process.**

Self-test

1. What does strict liability change about proving intent?
2. Which violation covers refusing a sample?
3. Why can an athlete need a TUE even when a medicine is medically necessary?
4. Which classes are S1, S3 and S9?

Sources

wada-code-2021; wada-prohibited-list-2026; wada-istue-2023; nada-india-rules-2021; nada-india-official-2026; nada-india-tue-2026; nada-india-abp-2026.

A2 — Major International Championships

Event fingerprint chart

Event	First / origin cue	Current high-yield cue
Commonwealth Games	1930, Hamilton; originally British Empire Games	Glasgow 2026 was the 23rd edition, 23 July–2 August, with a 10-sport programme and integrated Para sport; Amdavad is confirmed for 2030 centenary Games
Fédération Internationale de Football Association (FIFA) World Cup	1930, Uruguay; first champion Uruguay	2026: Canada–Mexico–United States; expanded 48-team tournament; keep format facts separate from historical winners
International Cricket Council (ICC) Cricket World Cup	One-day international (ODI) World Cup began in 1975	Distinguish the ODI World Cup, Twenty20 (T20) World Cup and World Test Championship; they are different competitions and formats
World Athletics Championships	First edition 1983, Helsinki	Tokyo 2025 was the 20th edition; athletics world championships are distinct from the Olympic athletics programme
World Aquatics Championships	First edition 1973, Belgrade	Aquatics includes swimming, diving, water polo, artistic swimming, open water and high diving; do not call it only a swimming meet
International Basketball Federation (FIBA) Basketball World Cup	First edition 1950, Argentina	2027 edition: Doha, Qatar; 32-team event

Host–format–governing body method

When reading an event question:

1. "Who governs?" asks for the federation or organisation.
2. "Where and when?" asks for the edition and host.
3. "What format?" asks which version of the competition is meant.
4. "Which was first?" requires the official history, not a rough memory.

Frequently confused pairs

- Olympic Games ≠ World Championships: Olympic cycle and federation championships have different histories and qualification systems.
- Commonwealth Games ≠ Commonwealth Youth Games: different age/edition structures.
- FIFA World Cup ≠ Club World Cup: national teams versus clubs.
- ICC ODI World Cup ≠ ICC T20 World Cup: different formats.
- World Aquatics Championships ≠ Olympic swimming: same sport family, different event.
- World Athletics Championships ≠ Diamond League: championship versus annual circuit meetings.

Easy ways to remember it

- Make a three-column card for each event: **first edition, governing body, current or next host**.
- Match the organisation to the sport in plain language: **FIFA is football, ICC is cricket, FIBA is basketball, World Athletics is track and field, and World Aquatics is water sports**.

Current-fact rule

These notes use the official snapshot available **2026-08-21**. A future host, completed Games result, sport programme or qualification format can change; check the organiser's page if the notification is updated.

Self-test

1. Which event began at Hamilton in 1930?
2. What is the difference between an international federation championship and the Olympics?
3. Which body is associated with basketball's World Cup?
4. Name two disciplines included in World Aquatics beyond pool swimming.

Sources

sai-notification-2026; commonwealth-sport-history-2026; commonwealth-sport-glasgow-2026; commonwealth-sport-amdavad-2030; fifa-world-cup-history; fifa-world-cup-2026; icc-world-cups-2026; world-athletics-champs-history-2025; world-athletics-tokyo-2025; world-aquatics-championships-history; world-aquatics-singapore-2025; fiba-basketball-world-cup-history; fiba-basketball-world-cup-2027.

A2 — Sports Governance, Ethics & Safeguarding

Safe-sport response tree

If you observe or receive a safeguarding concern:

1. If there is immediate danger, protect the athlete and call for emergency help.
2. Listen calmly without leading the athlete or promising absolute secrecy.
3. Record factual observations and preserve relevant evidence.
4. Report through the organisation's designated safeguarding channel.
5. Support the athlete, prevent retaliation and document follow-up.

Do not investigate alone, confront the alleged person in a way that increases risk, or circulate details on a team chat.

Governance map

Layer	Main question
Olympic Charter / International Olympic Committee (IOC)	What principles govern the Olympic Movement?
Ministry of Youth Affairs and Sports (MYAS) / national framework	What national policy and recognition framework applies?
Sports Authority of India (SAI)	How are athlete-development and support programmes implemented?
Indian Olympic Association (IOA)	How is India represented in the Olympic Movement?
National Sports Federation (NSF)	Who administers a particular sport, its selection and technical pathway?
Ethics/safeguarding bodies	How are welfare, conflicts, complaints and safe participation protected?

National Sports Governance Act: exam caution

The **National Sports Governance Act, 2025** creates a statutory framework for national sports bodies, ethics, athlete representation, dispute resolution and accountable governance. The commencement clause matters: an Act can be enacted but provisions may require a government commencement notification. For current-status MCQs, use the official notification rather than assuming every provision is already operational.

Ethics flash list

- **Autonomy:** respect the athlete's informed choice.
- **Beneficence/non-maleficence:** promote welfare; avoid preventable harm.
- **Justice:** fair access and non-discrimination.
- **Confidentiality:** share minimum necessary information with legitimate recipients.
- **Competence:** work within scope; refer when required.
- **Conflict of interest:** disclose and manage competing incentives.
- **Integrity:** accurate records, honest reporting, no manipulation for selection.

Safeguarding vocabulary

Safeguarding covers prevention and response to physical, sexual and emotional abuse, neglect, bullying, harassment, discrimination, exploitation and retaliation. A professional's job is not only to recognise a visible injury; abrupt behaviour change, unexplained injuries, fear of a staff member, secrecy or coercive control can be warning signs.

Sexual Harassment of Women at Workplace (POSH) exam cues

Under India's POSH framework, remember: an aggrieved woman can complain about sexual harassment at a workplace; the Internal Committee handles complaints where applicable; confidentiality protects the complaint/inquiry; employers have prevention and redressal duties. Time limits and procedure should be read from the current Act/policy when a question asks for an exact number.

Easy ways to remember it

- In an athlete-welfare concern, **listen, report through the correct channel, record facts and support the athlete.**
- In an ethics question, choose the response that protects **consent, autonomy, respect and fair treatment.**
- For governance, think from broad to local: **the Olympic Charter, the Ministry, the national sports organisations, the team policy and finally the athlete.**

Self-test

1. What is the first priority when a safeguarding concern presents immediate danger?
2. Why is "confidential" different from "secret from everyone"?
3. Which body is sport-specific: IOA or NSF?
4. Why must commencement status be checked for a new Act?

Sources

sai-notification-2026 ; a2-national-sports-governance-act-2025 ; myas-national-sports-code-2011 ; national-sports-code-2011 ; a2-ioc-safe-sport-2025 ; a2-ioc-safeguarding-toolkit-2017 ; a2-draw-the-line-ioc ; a2-sai-athlete-wellbeing-2026 ; a2-india-posh-act-2013 ; world-physiotherapy-ethics-2023 ; world-physiotherapy-consent-2023 .

B — Biomechanics & Movement Analysis

Kinematics versus kinetics

Kinematics: what moves? position, displacement, velocity, acceleration, angle
Kinetics: why does it move? force, torque, impulse, work, power, momentum

Formula strip

$v = \text{displacement} / \text{time}$
 $F = m \times a$
 $p = m \times v$
 $\text{work} = \text{force} \times \text{displacement}$
 $\text{pressure} = \text{force} / \text{area}$

$a = \Delta v / \Delta t$
 $\text{torque } (\tau) = \text{force} \times \text{perpendicular moment arm}$
 $\text{impulse} = \text{force} \times \text{time} = \text{change in momentum}$
 $\text{power} = \text{work} / \text{time} = \text{force} \times \text{velocity}$
 $\text{mechanical advantage} = \text{output force} / \text{input force}$

Vector questions require **magnitude and direction**. Ground-reaction force is the external force from the ground; it is not automatically the same as muscular force.

Levers

Class	Order	Common body example
First	F-axis-R	Head/neck extension
Second	axis-R-F	Heel raise
Third	axis-F-R	Biceps flexing the elbow

Most human limbs favour speed and range of motion over force advantage. Increasing a moment arm increases torque for the same muscle force, but changes movement speed and joint loading.

Stability and alignment

Stability increases when: the centre of mass stays over the base of support, the base is wider, the centre of mass is lower, friction is adequate, and the external disturbance is manageable.

Dynamic alignment questions often use **hip-knee-foot**: poor hip control can contribute to dynamic knee valgus, but the cause must be tested rather than assumed.

High-yield joint mechanics

- **Scapulohumeral rhythm**: a commonly taught overall 2:1 glenohumeral-to-scapulothoracic contribution during full elevation; actual ratios vary by range, task and person.
- **Knee screw-home**: terminal extension couples tibial external rotation relative to femur in open chain (or femoral internal rotation relative to tibia in closed chain); popliteus "unlocks" the knee.
- **Windlass mechanism**: toe dorsiflexion tensions plantar fascia, raises the medial arch and links foot stiffness to propulsion.
- **Subtalar pronation/supination**: triplanar motions; avoid reducing them to one plane.

Walking sequence

Part of walking	Sub-phases in order
Stance	Initial contact, loading response, mid-stance, terminal stance, pre-swing
Swing	Initial swing, mid-swing, terminal swing

Initial swing produces much of peak knee flexion for foot clearance. Terminal swing prepares limb positioning for the next initial contact. Interpret gait by timing, joint motion, moments, muscle action and task speed together.

Sport movement cues

- Jump landing: absorb force through coordinated hip/knee/ankle flexion; trunk position, knee alignment, stiffness and landing strategy interact.
- Throwing phases are **wind-up, cocking, acceleration, deceleration and follow-through**; deceleration is a major load-management issue.
- Sprinting: acceleration and maximal velocity differ in trunk angle, ground-contact time, step length/frequency and horizontal force orientation.

Easy ways to remember it

- **Kinematics describes the movement; kinetics explains the forces that cause or control it.**
- For levers, identify the middle item: first-class has the joint axis in the middle, second-class has the resistance in the middle, and third-class has the effort in the middle.
- Learn the walking sequence as words: **initial contact, loading response, mid-stance, terminal stance, pre-swing, initial swing, mid-swing, terminal swing.**

Self-test

1. What is the difference between torque and force?
2. Which lever class is elbow flexion by biceps commonly taught as?
3. What does the windlass mechanism do during toe dorsiflexion?
4. Why is a single 2-D video angle insufficient to diagnose a complex 3-D movement problem?

Sources

sai-notification-2026; basic-biomechanics-hall; neumann-kinesiology; perry-burnfield-gait-analysis; nsca-essentials-5; brukner-khan-clinical-sports-medicine.

B — Human Anatomy & Kinesiology

Planes, axes and movement

Plane	Axis	Typical movement
Sagittal	Mediolateral	Flexion/extension
Frontal	Anteroposterior	Abduction/adduction
Transverse	Vertical/longitudinal	Rotation

Rotator-cuff and shoulder anchors

Muscle	Action / nerve cue
Supraspinatus	Initiates abduction; suprascapular nerve
Infraspinatus	External rotation; suprascapular nerve
Teres minor	External rotation; axillary nerve
Subscapularis	Internal rotation; upper/lower subscapular nerves
Deltoid	Major abductor after initiation; axillary nerve
Serratus anterior	Scapular protraction/upward rotation; long thoracic nerve
Trapezius	Scapular elevation/retraction/upward rotation; spinal accessory nerve plus cervical contributions

The four rotator-cuff muscles are **supraspinatus**, **infraspinatus**, **teres minor** and **subscapularis**. Learn the full names because each muscle has a different action and nerve supply.

Lower-limb anchors

Region	High-yield cue
Hip	Gluteus medius stabilises pelvis in single-leg stance; superior gluteal nerve
Knee	Quadriceps extends knee; femoral nerve. Hamstrings flex knee/extend hip; tibial division of sciatic nerve predominates
Ankle	Tibialis anterior dorsiflexes; deep fibular nerve. Gastrocnemius/soleus plantarflex; tibial nerve
Foot	Tibialis posterior supports the medial arch; tibial nerve. Fibularis longus everts and supports the transverse arch; superficial fibular nerve

Peripheral nerves and quick neurological screen

Root/nerve level	Typical sensory or reflex cue
C5	Lateral upper arm; biceps reflex contribution
C6	Thumb/lateral forearm; brachioradialis contribution
C7	Middle finger; triceps contribution

Root/nerve level	Typical sensory or reflex cue
C8	Little finger/medial hand
T1	Medial forearm
L4	Medial leg; patellar reflex contribution
L5	Dorsum of foot/great toe; great-toe extension myotome
S1	Lateral foot; Achilles reflex contribution

Dermatomes, myotomes and reflexes are screening patterns, not isolated proof of a lesion. Compare sides and integrate pain, strength, sensation and neural tension.

Arthrokinematics

Convex-on-concave: roll and glide opposite directions
 Concave-on-convex: roll and glide same direction

Open kinetic chain: distal segment free; closed kinetic chain: distal segment fixed/loaded and multiple joints share demand. Neither is universally "safe" or "unsafe"; choose by stage, tissue, load and task.

Kinesiology reasoning

To analyse a movement, ask these questions in order: **Where is the joint? Which way does the muscle pull? How long is its moment arm? How much torque does it create? What movement or control does that produce?**

Easy ways to remember it

- Learn the rotator cuff as four complete names, not as an unexplained letter group.
- For a quick nerve check: the **femoral nerve helps knee extension, the axillary nerve supplies the deltoid, the tibial nerve helps plantarflexion, and the deep fibular nerve helps dorsiflexion.**
- Sensory landmarks: **C5 shoulder, C6 thumb, C7 middle finger, C8 little finger; L4 medial knee/leg, L5 great toe, S1 lateral foot.**
- When asked about a muscle, answer in four parts: **action, position, nerve and sporting task.**

Self-test

1. Which muscle initiates shoulder abduction in the classic 0–15° teaching point?
2. What is the difference between a dermatome and a myotome?
3. What happens to roll and glide in convex-on-concave movement?
4. Why can closed-chain exercise change joint loading at several joints at once?

Sources

sai-notification-2026; openstax-anatomy-physiology-2e; neumann-kinesiology; basic-biomechanics-hall; shacklock-clinical-neurodynamics; osullivan-rehabilitation; magee-orthopaedic-assessment.

B — Human Physiology & Exercise Physiology

Energy systems

System	Rate/capacity	Typical use cue
Adenosine triphosphate–phosphocreatine (ATP–PCr)	Highest rate, low capacity	Immediate maximal efforts
Anaerobic glycolysis	High rate, moderate capacity	Hard efforts lasting longer than the immediate burst; metabolites accumulate
Oxidative phosphorylation	Lower rate, high capacity	Sustained work and recovery between efforts

All three operate together. “Aerobic versus anaerobic” is a relative-contribution question, not an on/off switch.

Muscle physiology

Muscle contraction follows this sequence:

1. A nerve signal reaches the muscle.
2. Acetylcholine is released at the neuromuscular junction.
3. The muscle-fibre electrical signal travels along the fibre.
4. Calcium is released.
5. Calcium moves troponin so actin and myosin can form cross-bridges.
6. The cross-bridges cycle using adenosine triphosphate.

- **Henneman size principle:** smaller, lower-threshold motor units are recruited before larger, higher-threshold units as demand rises.
- Type I fibres: oxidative, fatigue-resistant, lower maximal force/power.
- Type II fibres: greater force/power and faster shortening, generally more fatigable; training and task alter phenotype and function.
- Isometric = tension without visible joint movement; concentric = muscle shortens; eccentric = muscle lengthens under tension.

Cardiovascular equations

Cardiac output = heart rate $\times$ stroke volume
Oxygen uptake = cardiac output $\times$ (arterial oxygen – venous oxygen difference) [Fick principle]
Mean arterial pressure $\approx$ diastolic pressure + $\frac{1}{3}$ (systolic pressure – diastolic pressure)

Acute exercise: heart rate, stroke volume (until high intensities in some conditions), cardiac output, ventilation and systolic blood pressure generally rise; blood flow is redistributed according to demand and thermoregulation.

Training adaptations can include lower submaximal heart rate, increased stroke volume, improved oxygen extraction, plasma-volume change and improved economy—depending on programme and athlete.

Threshold and fatigue

- Lactate threshold/ventilatory threshold marks a change in the relationship between work and metabolic/ventilatory response; exact definitions vary by test.
- Fatigue is multifactorial: substrate availability, metabolite/ion changes, neuromuscular drive, thermoregulation, central factors, pain and motivation may contribute.
- A blood biomarker alone rarely explains performance.

Thermoregulation and altitude

In heat, the body sends more blood to the skin and produces sweat; evaporation of that sweat removes heat. High humidity reduces evaporation and therefore increases heat strain. At altitude, lower oxygen pressure increases breathing and cardiovascular stress.

Heat acclimatisation is progressive exposure with monitoring. Altitude illness is clinical; headache plus nausea, dizziness, sleep disturbance or deteriorating neurological/respiratory symptoms should not be treated as ordinary "fitness adaptation."

Recovery physiology

Recovery restores homeostasis and supports supercompensation, but the response depends on dose, sleep, food, illness, psychological stress and tissue injury. Do not use "more inflammation is always bad" or "more soreness means more adaptation" as rules.

Easy ways to remember it

- Think of the energy systems as **fast, hard and long**: phosphocreatine is fastest, glycolysis supports hard efforts, and oxidative metabolism supports long efforts.
- Cardiac output means **how much blood the heart pumps each minute**, so it equals heart rate multiplied by stroke volume. Oxygen uptake equals cardiac output multiplied by the difference in oxygen content between arterial and venous blood.
- Muscle activation follows this order: **nerve signal, acetylcholine, calcium, troponin and finally the myosin cross-bridge**.

Self-test

1. Which system has the greatest power, and which has the greatest capacity?
2. What does the Fick equation tell you conceptually?
3. What is the size principle?
4. Why does high humidity increase heat strain even at the same air temperature?

Sources

sai-notification-2026; physiology-sport-exercise-kenney; nsca-essentials-5; openstax-anatomy-physiology-2e; acsm-getp-12; brukner-khan-clinical-sports-medicine-vol2.

B — Electrotherapy & Electrodiagnosis

Dose vocabulary

Amplitude/intensity = how strong
Pulse duration/width = how long each pulse lasts
Frequency = pulses per second (hertz, Hz)
Duty cycle = on-time / total cycle time
Polarity = direction of current in relevant systems

Always pair modality with **indication, dose, contraindication, skin sensation, tissue stage and outcome measure**.

Modality quick chart

Modality	Exam anchor
Conventional transcutaneous electrical nerve stimulation (TENS)	Higher frequency, comfortable sensory intensity; gate-control framing is common
Low-frequency transcutaneous electrical nerve stimulation	Lower frequency, stronger/motor-level tolerance; endogenous opioid mechanisms are commonly discussed
Neuromuscular electrical stimulation (NMES)	Evokes contraction to support activation/strength; electrode placement and voluntary effort matter
Interferential current (IFC)	Two medium-frequency currents intersect to create a beat/interference frequency; do not confuse carrier with beat frequency
Therapeutic ultrasound	1 megahertz (MHz) is generally deeper; 3 megahertz is generally more superficial. Continuous = more thermal; pulsed = lower average thermal dose
Short-wave diathermy (SWD)	Radiofrequency heating; capacitive/inductive coupling and metal/implant safety are key
Cryotherapy	Reduces tissue temperature and can alter pain/nerve conduction; protect skin and circulation
Photobiomodulation	Dose, wavelength, energy density and target tissue matter; "laser" is not one uniform treatment
Extracorporeal shock-wave therapy (ESWT)	Radial and focused systems differ in physics and depth; use condition-specific evidence

Strength–duration concepts

Rheobase = minimum current amplitude for a very long pulse
Chronaxie = pulse duration needed at twice rheobase to excite tissue

Denervated muscle requires different pulse characteristics from innervated muscle. Interpret a strength–duration curve with clinical examination and nerve status; do not diagnose from one curve alone.

Safety screen

- Pacemaker/implanted electronic device: modality-specific caution/contraindication; follow device and local guidance.
- Avoid current/heat/light over active malignancy, uncontrolled bleeding, impaired sensation/circulation or areas where skin/tissue safety cannot be monitored.
- Consider pregnancy, open epiphyseal plates, infection, thrombosis, metal/implants and medication effects according to modality.
- Never use electrotherapy to postpone assessment of a red flag or to mask worsening neurological/vascular symptoms.

Evidence-based choice

Choose a modality in five steps: **identify the clinical problem, define the desired outcome, choose the best-supported option, select a safe dose, and reassess whether a meaningful change occurred.**

The presence of a physiological effect does not prove a clinically meaningful outcome. For chronic nonspecific low-back pain, do not assume therapeutic ultrasound is essential merely because it produces tissue heating.

Easy ways to remember it

- When choosing a current, check four settings in plain language: **how strong it is, how long each pulse lasts, how often pulses repeat, and how much of each cycle is switched on.**
- Therapeutic ultrasound: **1 megahertz is generally deeper and 3 megahertz is generally more superficial;** continuous treatment heats more than pulsed treatment.
- Electrodiagnosis: **rheobase is the current level; chronaxie is the time needed.**

Self-test

1. What is the difference between carrier and beat frequency in an interferential current?
2. Which ultrasound frequency is generally used for deeper tissue: 1 megahertz or 3 megahertz?
3. Define rheobase and chronaxie.
4. Why is a physiological response insufficient evidence of treatment effectiveness?

Sources

sai-notification-2026; watson-electrotherapy; claytons-electrotherapy; acsm-getp-12; magee-orthopaedic-assessment.

B — Exercise Therapy & Rehabilitation

Rehabilitation staircase

Rehabilitation staircase

Progress through these stages when the athlete meets the relevant criteria:

- 1. Protect the tissue and calm unnecessary aggravation.
- 2. Restore range of motion and muscle activation.
- 3. Build strength and general capacity.
- 4. Add balance, power, speed and change-of-direction work.
- 5. Practise the sport task.
- 6. Expose the athlete to unrestricted performance demands.

Progress by **response and criteria**, not by calendar alone.

Exercise principles

Specificity · overload · progression · individualisation · reversibility · variation · recovery.

For every exercise specify: position, load, range, speed, sets/repetitions, rest, pain response, quality criterion and progression/regression.

Contraction and loading

Mode	Useful cue
Isometric	Tension at a fixed joint angle; useful when movement is irritable or for activation, but angle-specific
Concentric	Muscle shortens while producing force
Eccentric	Muscle lengthens under load; high force and tissue-loading stimulus; dose progressively
Isokinetic	Constant angular speed with accommodating resistance; specialised equipment

For tendinopathy, progressive tendon loading may include isometrics, isotonic/eccentric or heavy-slow resistance and sport exposure; the best choice depends on stage, symptoms and capacity.

Flexibility and proprioceptive neuromuscular facilitation

Method	Main cue
Static	Held position; low ballistic demand
Dynamic	Controlled movement through range; useful in preparation
Ballistic	Bouncing/rapid end-range; higher demand and context-sensitive
Hold-relax	Isometric contraction followed by relaxation and increased range
Contract-relax	Concentric/isotonic contraction followed by relaxation and increased range
Agonist contraction	Antagonist relaxation facilitated by active agonist movement

Open versus closed chain

Open: distal segment free; isolate or bias a muscle/joint angle
Closed: distal segment fixed; co-contraction and distributed loading

Choose based on tissue protection, joint shear/compression, task demand and athlete tolerance—not labels alone.

Maitland grades

I small amplitude, beginning of range	pain modulation
II large amplitude, within available range	pain / movement tolerance
III large amplitude, into resistance	mobility
IV small amplitude, at end range	mobility
V small, fast thrust at end range	manipulation by trained clinician

The grade is not the whole treatment: direction, examination findings, irritability, contraindications and reassessment decide appropriateness.

Proprioception, plyometrics, hydrotherapy

- Proprioception: progress from stable to unstable, predictable to perturbed, two-leg to one-leg, and slow to fast tasks.
- Plyometrics: eccentric braking followed by rapid concentric action; require adequate strength, landing control and tissue readiness.
- Aquatic therapy unloads body weight and changes resistance; water depth, buoyancy, drag and temperature alter the dose.

Easy ways to remember it

- Progress in this order: **restore movement, restore activation, build strength, then practise the sport task.**
- Maitland grades I and II are usually used for gentle pain or movement tolerance; grades III and IV emphasise mobility; grade V is a thrust performed by an appropriately trained clinician.
- Increase difficulty step by step: **range, load, speed, complexity, then unpredictable sport demands.**

Self-test

1. What criterion should govern progression from strengthening to plyometrics?
2. Distinguish eccentric from concentric contraction.
3. Why can closed-chain exercise distribute load across several joints?
4. Which Maitland grades are classically associated with end-range mobility?

Sources

sai-notification-2026; kisner-therapeutic-exercise; maitland-peripheral-manipulation; sports-injury-prevention-rehabilitation-joyce-lewindon; brukner-khan-clinical-sports-medicine; nsca-essentials-5; osullivan-rehabilitation.

B — Assessment, Diagnosis & Clinical Reasoning

Examination sequence

Examine the athlete in this order:

1. Take the subjective history and screen for red flags.
2. Observe the athlete and assess a relevant functional task.
3. Palpate, measure active and passive range of motion, and test resisted movement.
4. Perform manual muscle, neurological and circulation checks.
5. Add special tests and a validated outcome measure.
6. Form a working diagnosis and differential diagnosis.
7. Make a plan and reassess the response.

Four-part clinical note

Subjective: history, symptoms, mechanism, goals, 24-hour response. **Objective:** measured findings and test conditions. **Assessment:** interpretation, differential, irritability, impairments and functional meaning. **Plan:** intervention, education, load, referral/escalation and review date.

Diagnostic accuracy

Sensitivity = $\text{true positives} / (\text{true positives} + \text{false negatives})$
Specificity = $\text{true negatives} / (\text{true negatives} + \text{false positives})$
Positive likelihood ratio = $\text{sensitivity} / (1 - \text{specificity})$
Negative likelihood ratio = $(1 - \text{sensitivity}) / \text{specificity}$

Use a high-sensitivity test to help rule out when negative and a high-specificity test to help rule in when positive. Test accuracy varies by examiner, population, disease stage and reference standard.

Common exam associations (not stand-alone diagnoses)

Test	Common association
Lachman	Anterior cruciate ligament (ACL) integrity; often taught as a sensitive ACL test
Pivot shift	Rotational ACL instability; often taught as a more specific ACL test
McMurray/Thessaly	Meniscal involvement; interpret with history and other findings
Anterior ankle drawer	Anterior talofibular ligament (ATFL)/lateral ankle instability
Talar tilt	Calcaneofibular ligament (CFL)/lateral ligament stress
Thompson/Simmonds	Achilles tendon rupture screen
Apprehension/relocation	Anterior shoulder instability
Neer/Hawkins-Kennedy	Shoulder pain/impingement provocation; not a single-tissue diagnosis
Speed/Yergason	Biceps-related provocation patterns
Flexion–abduction–external rotation (FABER)	Hip/sacroiliac-region provocation; location of symptoms matters
Straight-leg raise/slump	Neural mechanosensitivity/radicular pattern; structural differentiation matters

Manual muscle testing and range of motion

- 0 no visible/palpable contraction
- 1 trace contraction
- 2 full range in gravity-eliminated position
- 3 full range against gravity
- 4 moderate resistance
- 5 normal/maximal resistance for the test position

End-feel: hard (bone), firm (capsule/muscle), soft (tissue approximation), empty (pain stops movement), or abnormal/springy depending on context. Document pain, quality, range, side and position.

Outcome measures

Domain	Examples
Pain	Numeric Pain Rating Scale / Visual Analogue Scale
Lower limb function	Lower Extremity Functional Scale; Knee injury and Osteoarthritis Outcome Score; International Knee Documentation Committee score
Upper limb	Disabilities of the Arm, Shoulder and Hand / QuickDASH
Tendon	Victorian Institute of Sport Assessment–Achilles / –Patellar
Psychological readiness	Anterior Cruciate Ligament–Return to Sport after Injury scale and similar sport-specific tools

Red flags and referral

Progressive neurological deficit, fracture/dislocation concern, vascular compromise, infection/systemic illness, unexplained night pain/weight loss, cauda-equina features, severe mechanism or deteriorating consciousness override routine exercise testing.

Easy ways to remember it

- Examine in this order: **history, observation, function, nerves and circulation, then special tests.**
- **Rule out danger before naming the injured tissue.**
- For every test ask three simple questions: **What structure or movement is stressed? What is the best comparison or reference? How reliable is this test in this situation?**

Self-test

1. Which test characteristic is most useful to rule out when negative?
2. What belongs in the assessment part of a clinical note?
3. Why is a positive provocation test not automatically a tissue diagnosis?
4. What does an empty end-feel suggest about why movement stopped?

Sources

sai-notification-2026; magee-orthopaedic-assessment; konin-special-tests-orthopedic-examination; shacklock-clinical-neurodynamics; world-physiotherapy-ebp-2023; brukner-khan-clinical-sports-medicine.

B — Sports Physiotherapy, Injury Management & Return to Play

Acute soft-tissue framework

Early care focuses on **protection, elevation, avoiding unnecessary anti-inflammatory use, compression and education**. Later care focuses on **progressive loading, a constructive recovery outlook, improving blood flow and exercise**.

The key idea is not to immobilise everything or to eliminate all symptoms; protect serious tissue, control unnecessary aggravation, educate, then progressively reload capacity.

Injury-specific anchors

Problem	High-yield cue
Lateral ankle sprain	The anterior talofibular ligament (ATFL) is most commonly involved, then the calcaneofibular ligament (CFL); restore range of motion, strength, balance and sport-specific loading
Syndesmotic sprain	"High ankle" pattern; often slower and different from uncomplicated lateral sprain
Hamstring strain	Rebuild isometric strength, lengthened loading, high-speed running exposure and sport demands progressively
Anterior cruciate ligament (ACL) reconstruction	Criteria-based strength, range of motion, effusion, neuromuscular control, hopping/running and psychological readiness; time alone is insufficient
Groin pain	Doha taxonomy: adductor-related, iliopsoas-related, inguinal-related and pubic-related; examination is symptom-location and resisted/palpation based
Stress injury	Respect bone-risk site, load history and imaging/medical referral; pain suppression is not healing
Tendinopathy	Progressive tendon capacity and energy-storage loading; avoid promising a single universal protocol

Return-to-sport continuum

Stage	Meaning
Return to participation	The athlete can take part with a modified load
Return to sport	The athlete can complete full practice demands
Return to performance	The athlete can tolerate competition-level demands

Exit criteria

- Symptoms/effusion controlled and stable.
- Full or functionally adequate range of motion; strength and rate-of-force/power appropriate to sport.
- Limb symmetry interpreted alongside absolute capacity and task quality; **a limb symmetry index (LSI) of at least 90% is a common screen, not a universal clearance rule**.
- Hop/change-of-direction/running tests, repeated exposure and sport-specific workload tolerated.
- Psychological readiness and athlete shared decision included.
- Medical, physiotherapy, coaching and athlete communication aligned.

Prevention cues

- Neuromuscular warm-ups such as FIFA 11+ are programmes, not one exercise.
- Nordic hamstring exercise targets eccentric hamstring capacity; adherence and progression matter.
- Copenhagen adduction targets adductor strength; dose to capacity.
- Screening identifies modifiable risk or baseline, but a screen alone does not predict an individual injury with certainty.

Easy ways to remember it

- Early injury care means **protect and educate**; later care means **restore load and exercise capacity**.
- Return to sport asks three plain questions: **Can the body do the task? Will the athlete tolerate the demand repeatedly? Has the athlete and team agreed that returning now is sensible?**
- Groin categories in the Doha agreement are **adductor-related, iliopsoas-related, inguinal-related and pubic-related**.

Self-test

1. Why is a time-based ACL clearance incomplete?
2. What is the difference between return to participation and return to performance?
3. Name the four Doha groin categories.
4. Why should a limb symmetry index not be used as the only return-to-sport criterion?

Sources

sai-notification-2026; brukner-khan-clinical-sports-medicine; sports-injury-prevention-rehabilitation-joyce-lewindon; aspetar-acl-cpg-2023; london-hamstring-consensus-2023; doha-groin-consensus-2015; apta-lateral-ankle-cpg-2021; ioc-load-management-consensus-2016; bern-return-to-sport-2016.

B — Sports Emergencies, On-Field Management & Travel

Emergency action plan

An emergency action plan should cover five tasks: **prevent problems where possible, plan and rehearse roles/equipment/access, recognise danger, respond with the primary survey and emergency activation, then document and improve the plan afterwards.**

Primary survey

The primary survey checks, in order, **catastrophic bleeding, airway with cervical-spine awareness, breathing, circulation, disability, and exposure to the environment.** Treat life threats as found; use local emergency protocols and qualified responders.

- Attach an **automated external defibrillator (AED)** and follow its prompts as soon as it is available for a suspected cardiac arrest; minimise pauses in high-quality **cardiopulmonary resuscitation (CPR)**.
- Severe bleeding: direct pressure/tourniquet according to training and site protocol.
- Suspected spinal injury: manual stabilisation, airway/breathing priorities, coordinated movement and appropriate equipment; do not delay life-saving care for an ideal transfer technique.

Concussion / Sport Concussion Assessment Tool 6 (SCAT6)

For a suspected concussion:

1. If a red flag is present, arrange emergency medical assessment or transport.
2. If no red flag is present, remove the athlete from play and assess using more than one type of information.
3. Do not allow same-day return to play.
4. Use a symptom-limited, gradual return-to-learn and return-to-sport pathway after appropriate medical assessment.

Red flags include deteriorating consciousness, repeated vomiting, seizure, severe/worsening headache, neck pain/tenderness, double vision, focal weakness/tingling, unusual behaviour or increasing confusion. The Sport Concussion Assessment Tool 6 supports acute assessment; it is not a stand-alone diagnosis and normal findings do not erase clinical concern.

Heat and cold emergencies

Condition	Priority
Exertional heat stroke	Rapid recognition and cold-water immersion/cooling immediately ; do not delay cooling for transport when safe protocol permits
Heat exhaustion	Remove from heat, cool, assess hydration/medical status; exclude heat stroke
Hypothermia	Remove wet exposure, gentle handling, rewarm and assess airway/circulation; severe cases need medical care
Altitude illness	Stop ascent; assess symptoms; descent/oxygen/medical care according to severity and protocol

Travel readiness

- Carry the emergency action plan, medical kit, emergency contacts, medication/therapeutic-use-exemption documents and a record of allergies/conditions with appropriate privacy controls.

- Use movement breaks during long travel; protect sleep, meal timing and hydration.
- After time-zone change, anchor local sleep/wake and light exposure; reduce high-risk intensity until alertness and recovery are acceptable.
- Infection-control, food/water safety, climate, altitude, venue access and insurance/evacuation plans belong in the travel risk assessment.

Easy ways to remember it

- On the field, look in this order: **catastrophic bleeding, airway, breathing, circulation, disability and exposure.**
- Suspected exertional heat stroke: **start cooling immediately and keep reassessing.**
- Suspected concussion: **look for red flags, remove the athlete from play, then begin a gradual rehabilitation pathway.**

Self-test

1. What takes priority over a detailed musculoskeletal examination on the field?
2. Name three concussion red flags.
3. What is the immediate priority in exertional heat stroke?
4. Why should a travel plan include emergency access and medication documentation?

Sources

sai-notification-2026; scat6-consensus-bjsm-2023; ioc-concussion-amsterdam-2022; ioc-manual-sports-injuries; magee-orthopaedic-assessment; brukner-khan-clinical-sports-medicine-vol2; physiology-sport-exercise-kenney; ioc-load-illness-2016; apta-lateral-ankle-cpg-2021.

B — Sports Medicine, Pharmacology, Nutrition & Research Basics

Clinical–clean-sport rule

For a medicine decision, first diagnose the problem and consider non-drug options. Then check the medicine's indication, route, dose, interactions and side effects. Check its anti-doping status and whether a therapeutic use exemption is needed. Document the decision and reassess the athlete.

Never assume a medicine is permitted because it is common, prescribed, inhaled, topical or bought without prescription. Check the current **World Anti-Doping Agency (WADA)** List and **National Anti-Doping Agency India (NADA India)** guidance.

High-yield pharmacology cues

- **Non-steroidal anti-inflammatory drugs (NSAIDs)** may affect gastrointestinal, renal and cardiovascular risk; they can mask symptoms and should not replace load modification or diagnosis.
- Glucocorticoid status depends on route and competition period; consult the current List and route-specific guidance.
- Opioids and sedating medicines can impair reaction, judgement and safe participation.
- Beta-2 agonists, stimulants, anabolic agents, peptide hormones and masking agents are frequent anti-doping categories.
- Document allergies, interactions, contraindications and the athlete's consent/understanding.

Nutrition for performance and rehabilitation

Need	Principle
Energy availability	Chronic mismatch between intake and expenditure can impair health and performance; screen for relative energy deficiency in sport (RED-S) risk
Carbohydrate	Supports high-intensity work and glycogen restoration; timing depends on training density
Protein	Distribute adequate high-quality protein; needs may rise during immobilisation/rehabilitation, but prescribe individually
Fat	Essential for energy, membranes and hormone-related functions; do not eliminate it
Hydration	Match duration, climate, sweat rate and sodium losses; avoid both severe dehydration and overdrinking
Supplements	Food first; check evidence, contamination risk, interaction, dose and third-party testing

Approximate energy values: carbohydrate **4 kcal/g**, protein **4 kcal/g**, fat **9 kcal/g**. Numbers such as protein g/kg/day are starting points, not universal prescriptions; consider injury, body size, total energy, renal/hepatic status and dietitian input.

Research design ladder

One useful evidence hierarchy is:

1. Systematic review or meta-analysis.
2. Randomised controlled trial.
3. Cohort study.
4. Case-control study.
5. Cross-sectional study.

6. Case series or case report.
7. Expert opinion.

The hierarchy is a shortcut, not a verdict: question, bias, precision, applicability and study quality matter.

Population–intervention–comparison–outcome (PICO) and statistics

P = population/problem I = intervention/exposure
C = comparator O = outcome

Relative risk (RR) = risk exposed / risk control
Absolute risk reduction (ARR) = risk control – risk exposed
Number needed to treat (NNT) = 1 / ARR
Odds ratio (OR) = odds exposed / odds control

- RCTs support causal intervention inference when well designed; observational studies are often essential when randomisation is unethical/impractical.
- A p-value is not effect size; read the confidence interval and clinical threshold.
- Sensitivity catches disease; specificity clears disease. Predictive values depend on prevalence.
- Bias types: selection, measurement/information, recall, confounding, publication and attrition bias.

Evidence-to-practice question

When applying research, ask: **Is the finding trustworthy? Is the effect important? Does it apply to this athlete? What are the benefits, harms, cost and athlete preferences?**

Easy ways to remember it

- Before believing a research result, ask four questions: **Who was studied? What was done? What was it compared with? What outcome was measured?**
- Research designs move from **summarising many studies, to randomised experiments, to following groups, to looking back at records, to describing cases.**
- For medicines, check **route, competition status, side effects and documentation.**

Self-test

1. Why can route change a drug's anti-doping status?
2. Distinguish ARR from NNT.
3. What four elements form PICO?
4. Why should statistical significance be paired with clinical importance?

Sources

sai-notification-2026; wada-prohibited-list-2026; wada-istue-2023; nada-india-rules-2021; clinical-sports-nutrition-burke; ioc-sports-nutrition-consensus; ioc-reds-2023; world-physiotherapy-ebp-2023; nsca-essentials-5.